

MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Trigonometric Equation

(English Medium)

EXERCISE (O-1)

Straight Objective Type

1. If $2 \tan^2 \theta = \sec^2 \theta$, then the general solution of θ -

- (A) $n\pi + \frac{\pi}{4}$ ($n \in \mathbb{I}$) (B) $n\pi - \frac{\pi}{4}$ ($n \in \mathbb{I}$) (C) $n\pi \pm \frac{\pi}{4}$ ($n \in \mathbb{I}$) (D) $2n\pi \pm \frac{\pi}{4}$ ($n \in \mathbb{I}$)

TE0001

2. If $\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = 3$, then the general solution of θ is -

- (A) $2n\pi \pm \pi/6$ (B) $n\pi \pm \pi/6$ (C) $2n\pi \pm \pi/3$ (D) $n\pi \pm \pi/3$

where $n \in \mathbb{I}$

TE0002

3. The general value of θ satisfying $\sin^2 \theta + \sin \theta = 2$ is -

- (A) $n\pi + (-1)^n \frac{\pi}{6}$; $n \in \mathbb{Z}$ (B) $2n\pi + \frac{\pi}{4}$; $n \in \mathbb{Z}$
(C) $n\pi + (-1)^n \frac{\pi}{2}$; $n \in \mathbb{Z}$ (D) $n\pi + (-1)^n \frac{\pi}{3}$; $n \in \mathbb{Z}$

TE0003

4. Let $A = \{\theta : \sin(\theta) = \tan(\theta)\}$ and $B = \{\theta : \cos(\theta) = 1\}$ be two sets. Then -

- (A) $A = B$ (B) $A \subset B$ and $B - A \neq \phi$
(C) $A \not\subset B$ (D) $B \not\subset A$

TE0004

5. The solution set of $(5 + 4 \cos \theta)(2 \cos \theta + 1) = 0$ in the interval $[0, 2\pi]$ is :

- (A) $\left\{\frac{\pi}{3}, \frac{2\pi}{3}\right\}$ (B) $\left\{\frac{\pi}{3}, \pi\right\}$ (C) $\left\{\frac{2\pi}{3}, \frac{4\pi}{3}\right\}$ (D) $\left\{\frac{2\pi}{3}, \frac{5\pi}{3}\right\}$

TE0005

6. The general solution of equation $4 \cos^2 x + 6 \sin^2 x = 5$ is -

- (A) $x = n\pi \pm \frac{\pi}{2}$ ($n \in \mathbb{I}$) (B) $x = n\pi \pm \frac{\pi}{4}$ ($n \in \mathbb{I}$)
(C) $x = n\pi \pm \frac{3\pi}{2}$ ($n \in \mathbb{I}$) (D) None of these

TE0006

7. If $\tan^2 \theta - (1 + \sqrt{3}) \tan \theta + \sqrt{3} = 0$, then the general value of θ is :

- (A) $n\pi + \frac{\pi}{4}, n\pi + \frac{\pi}{3}$ (B) $n\pi - \frac{\pi}{4}, n\pi + \frac{\pi}{3}$ (C) $n\pi + \frac{\pi}{4}, n\pi - \frac{\pi}{3}$ (D) $n\pi - \frac{\pi}{4}, n\pi - \frac{\pi}{3}$

where $n \in \mathbb{I}$

TE0007

8. The general solution of the equation $\tan^2 \alpha + 2\sqrt{3} \tan \alpha = 1$ is given by -

- (A) $\alpha = \frac{n\pi}{2} (n \in \mathbb{I})$ (B) $\alpha = (2n+1) \frac{\pi}{2} (n \in \mathbb{I})$
 (C) $\alpha = (6n+1) \frac{\pi}{12} (n \in \mathbb{I})$ (D) $\alpha = \frac{n\pi}{12} (n \in \mathbb{I})$

TE0008

9. The number of solutions of the equation $\sin 2x - 2\cos x + 4 \sin x = 4$ in the interval $[0, 5\pi]$ is -

- (A) 6 (B) 4 (C) 3 (D) 5

TE0009

10. The set of angles between 0 and 2π satisfying the equation $4 \cos^2 \theta - 2\sqrt{2} \cos \theta - 1 = 0$ is

- (A) $\left\{ \frac{\pi}{12}, \frac{5\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12} \right\}$ (B) $\left\{ \frac{\pi}{12}, \frac{7\pi}{12}, \frac{17\pi}{12}, \frac{23\pi}{12} \right\}$
 (C) $\left\{ \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{19\pi}{12} \right\}$ (D) $\left\{ \frac{\pi}{12}, \frac{7\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12} \right\}$

TE0010

11. If $\tan \theta + \tan 4\theta + \tan 7\theta = \tan \theta \tan 4\theta \tan 7\theta$, then $\theta =$

- (A) $\frac{n\pi}{4}$ (B) $\frac{n\pi}{7}$
 (C) $\frac{n\pi}{12}, n \neq (12m-6), n, m \in \mathbb{I}$ (D) $n\pi$

where $n \in \mathbb{I}$

TE0011

12. If $\frac{\tan 2\theta + \tan \theta}{1 - \tan \theta \tan 2\theta} = 0$, then the general value of θ is -

- (A) $n\pi; n \in \mathbb{I}$ (B) $\frac{n\pi}{3}; n \in \mathbb{I}$ (C) $\frac{n\pi}{4}$ (D) $\frac{n\pi}{6}; n \in \mathbb{I}$

where $n \in \mathbb{I}$

TE0012

13. The smallest positive angle satisfying the equation $1 + \cos 3x - 2\cos 2x = 0$, is equal to

- (A) 15° (B) 22.5° (C) 30° (D) 45°

TE0013

14. If $\tan \theta - \sqrt{2} \sec \theta = \sqrt{3}$, then the general solution of θ is -

- (A) $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{3}$ (B) $n\pi + (-1)^n \frac{\pi}{3} - \frac{\pi}{4}$
(C) $n\pi + (-1)^n \frac{\pi}{3} + \frac{\pi}{4}$ (D) $n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{3}$

where $n \in \mathbb{I}$

TE0014

15. Number of principal solution(s) of the equation $4 \cdot 16^{\sin^2 x} = 2^{6 \sin x}$ is

- (A) 1 (B) 2 (C) 3 (D) 4

TE0015

16. The most general values of x for which $\sin x + \cos x = \min_{a \in \mathbb{R}} \{1, a^2 - 4a + 6\}$ is given by -

- (A) $2n\pi$ (B) $2n\pi + \frac{\pi}{2}$ (C) $n\pi + (-1)^n \cdot \frac{\pi}{4} - \frac{\pi}{4}$ (D) None of these

where $n \in \mathbb{I}$

TE0016

17. If the equation $\sin^4 x - (k+2)\sin^2 x - (k+3) = 0$ has a solution then k must lie in the interval :

- (A) $(-4, -2)$ (B) $[-3, 2)$ (C) $(-4, -3)$ (D) $[-3, -2]$

TE0017

18. Number of values of x satisfying the equation $\log_2(\sin x) + \log_{1/2}(-\cos x) = 0$ in the interval $(-\pi, \pi]$ is equal to -

- (A) 0 (B) 1 (C) 2 (D) 3

TE0018

19. The equation $\sin x \cos x = 2$ has :

- (A) one solution (B) two solutions (C) infinite solutions (D) no solution

TE0019

20. The number of solutions of the equation $\tan^2 x - \sec^{10} x + 1 = 0$ in $(0, 10)$ is -

- (A) 3 (B) 6 (C) 10 (D) 11

TE0020

21. If $x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$, then the greatest positive solution of $1 + \sin^4 x = \cos^2 3x$ is -

- (A) π (B) 2π (C) $\frac{5\pi}{2}$ (D) none of these

TE0021

22. The number of solutions of the equation $\sin x = x^2 + x + 1$ is-

- (A) 0 (B) 1 (C) 2 (D) None

TE0022

23. **Statement-1:** If $\sin \frac{3x}{2} \cos \frac{5y}{3} = k^8 - 4k^4 + 5$, where $x, y \in \mathbb{R}$ then exactly four distinct real values of k are possible.

because

Statement-2: $\sin \frac{3x}{2}$ and $\cos \frac{5y}{3}$ both are less than or equal to one and greater than or equal to -1 .

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

TE0023

24. The number of solutions of the equation $2\cos\left(\frac{x}{2}\right) = 3^x + 3^{-x}$ is -

- (A) 1 (B) 2 (C) 3 (D) None

TE0024

25. The equation $2\cos^2\left(\frac{x}{2}\right)\sin^2 x = x^2 + x^{-2}$, $0 < x \leq \frac{\pi}{2}$ has

- (A) one real solutions (B) more than one real solutions
 (C) no real solution (D) none of the above

TE0025

26. If $0 \leq x \leq 3\pi$, $0 \leq y \leq 3\pi$ and $\cos x \cdot \sin y = 1$, then the possible number of values of the ordered pair (x, y) is -

- (A) 6 (B) 12 (C) 8 (D) 15

TE0026

27. Given $a^2 + 2a + \operatorname{cosec}^2 \left(\frac{\pi}{2} (a+x) \right) = 0$ then, which of the following holds good ?

$$(A) \ a = 1 ; \frac{x}{2} \in I$$

(B) $a = -1$; $\frac{x}{2} \in I$

(C) $a \in R ; x \in \phi$

(D) a, x are finite but not possible to find

TE0027

28. If the equation $\cot^4 x - 2 \operatorname{cosec}^2 x + a^2 = 0$ has atleast one solution then, sum of all possible integral values of 'a' is equal to

(A) 4

(B) 3

(C) 2

(D) 0

TE0028

29. The complete solution set of the inequality $\tan^2 x - 2\sqrt{2} \tan x + 1 \leq 0$ is -

$$(A) \quad n\pi + \frac{\pi}{8} \leq x \leq \frac{3\pi}{8} + n\pi, \quad n \in \mathbb{I}$$

(B) $n\pi + \frac{\pi}{4} \leq x \leq \frac{3\pi}{4} + n\pi, n \in \mathbb{I}$

(C) $n\pi + \frac{\pi}{16} \leq x \leq \frac{3\pi}{8} + n\pi, n \in \mathbb{I}$

(D) $n\pi + \frac{\pi}{3} \leq x \leq \frac{2\pi}{3} + n\pi, n \in \mathbb{I}$

TE0029

30. Number of integral solution(s) of the inequality $2\sin^2 x - 5\sin x + 2 > 0$ in $x \in [0, 2\pi]$, is -

(A) 3

(B) 4

(C) 5

(D) 6

TE0030

EXERCISE (O-2)

Multiple Correct Answer Type

1. The equation $2\sin \frac{x}{2} \cdot \cos^2 x + \sin^2 x = 2\sin \frac{x}{2} \cdot \sin^2 x + \cos^2 x$ has a root for which

(A) $\sin 2x = 1$

(B) $\sin 2x = -1$

$$(C) \cos x = \frac{1}{2}$$

(D) $\cos 2x = -\frac{1}{2}$

TE0031

2. $\sin^2 x - \cos 2x = 2 - \sin 2x$ if

(A) $x = n\pi/2, n \in \mathbb{I}$

(B) $\tan x = 3/2$

(C) $x = (2n + 1) \pi/2, n \in I$

(D) $x = n\pi + (-1)^n \sin^{-1}(2/3), n \in \mathbb{I}$

TE0032

- 3.** $4 \sin^4 x + \cos^4 x = 1$ if

(A) $x = n\pi$

$$(B) \ x = n\pi \pm \frac{1}{2} \cos^{-1} \left(\frac{1}{5} \right)$$

$$(C) \ x = \frac{n\pi}{2}$$

(D) none of these, ($n \in I$)

TE0033

4. Which of the following set of values of x satisfies the equation

$$2^{(2\sin^2 x - 3\sin x + 1)} + 2^{(2 - 2\sin^2 x + 3\sin x)} = 9 \text{ is -}$$

(A) $x = x\pi, n \in \mathbb{I}$

(B) $x = 2n\pi + \frac{\pi}{2}, n \in I$

(C) $x = n\pi \pm \frac{\pi}{6}, n \in I$

(D) $x = n\pi \pm \frac{\pi}{3}, n \in \mathbb{I}$

TE0034

5. $5 \sin^2 x + \sqrt{3} \sin x \cos x + 6 \cos^2 x = 5$ if

(A) $\tan x = -1/\sqrt{3}$

(B) $\sin x = 0$

(C) $x = n\pi + \pi/2, n \in \mathbb{I}$

(D) $x = n\pi + \pi/6, n \in \mathbb{I}$

TE0035

6. If sum of all the solution of the equation $\cot x + \operatorname{cosec} x + \sec x = \tan x$ in $[0, 2\pi]$ is $\frac{k\pi}{2}$, then the value of k is greater than

(A) 1

(B) 2

(C) 3

(D) 4

TE0036

7. The equation $\sin x + \cos (k + x) + \cos (k - x) = 2$ has real solution(s), then $\sin k$ can be

- (A) $\frac{-3}{4}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) $\frac{3}{4}$

TE0037

8. Which of the following equations have no solution ?

- (A) $2^{|x|} = \sin x^2$ (B) $3^{\left|\sin \sqrt{x}\right|} = |\cos x|$ (C) $x^2 = -\cos x$ (D) $3x^2 = 1 - 2\cos x$

TE0038

9. If $m > 0, n < 5, 0 < m + n < 10$ and $\frac{x^2 + 5}{2} = x - 2 \cos (m + nx)$ has atleast one real root, then

- (A) the greatest value of $(m + n)$ is 3π .
(B) the greatest value of $(m + n)$ is 2π .
(C) the least value of $(m + n)$ is π .
(D) the least value of $(m + n)$ is 2π .

TE0039

10. If $x^4 + 3\cos(ax^2 + bx + c) = 2(x^2 - 2)$ has two solutions with $a, b, c \in (2, 5)$ then

- (A) $a + b + c = \pi$ (B) $a - b + c = \pi$ (C) $a + b + c = 3\pi$ (D) $a - b + c = 3\pi$

TE0040

11. The equation $\sin^6 x + \cos^6 x = a^2$ has real solution if

- (A) $a \in (-1, 1)$ (B) $a \in \left(-1, -\frac{1}{2}\right)$ (C) $a \in \left(-1, \frac{1}{2}\right)$ (D) $a \in \left(\frac{1}{2}, 1\right)$

TE0101

12. If $\sin(x - y) = \cos(x + y) = 1/2$ then the values of x & y lying between 0 and π are given by :

- (A) $x = \pi/4$, $y = 3\pi/4$ (B) $x = \pi/4$, $y = \pi/12$
(C) $x = 5\pi/4$, $y = 5\pi/12$ (D) $x = 11\pi/12$, $y = 3\pi/4$

TE0102

13. $\sin x, \sin 2x, \sin 3x$ are in A.P if

- (A) $x = n\pi/2, n \in I$ (B) $x = n\pi, n \in I$ (C) $x = 2n\pi, n \in I$ (D) $x = (2n + 1)\pi, n \in I$

TE0103

14. $\cos 4x \cos 8x - \cos 5x \cos 9x = 0$ if

- (A) $\cos 12x = \cos 14x$ (B) $\sin 13x = 0$
(C) $\sin x = 0$ (D) $\cos x = 0$

TE0104

15. $\sin x - \cos^2 x - 1$ assumes the least value for the set of values of x given by :

- (A) $x = n\pi + (-1)^{n+1} (\pi/6)$, $n \in \mathbb{I}$
 (B) $x = n\pi + (-1)^n (\pi/6)$, $n \in \mathbb{I}$
 (C) $x = n\pi + (-1)^n (\pi/3)$, $n \in \mathbb{I}$
 (D) $x = n\pi - (-1)^n (\pi/6)$, $n \in \mathbb{I}$

TE0105

EXERCISE (O-3)**Linked Comprehension Type****Paragraph for Question 1 and 2**

Let $f(x) = \cos x + \sin x - 1$ and $g(x) = \sin 2x - 2$.

1. Number of solutions of the equation $f(x) = g(x)$ in $x \in \left[-\pi, \frac{7\pi}{2}\right]$ is equal to

(A) 4 (B) 5 (C) 6 (D) 8

TE0041

2. If the equation $f(x) = k$ has atleast one real solution then the number of possible integral values of k is equal to

(A) 1 (B) 2 (C) 3 (D) 4

TE0041

Paragraph for Question 3 to 5

Let $f(x) = \sin^2 x - (a - 1) \sin x + 2(a - 3)$

On the basis of above information, answer the following questions :

3. If $x \in [0, \pi]$ and $f(x) = 0$ has exactly one real root, then 'a' lies in

(A) (3, 5) (B) (2, 4) (C) (4, 5) (D) none of these

TE0042

4. If $f(x) = 0$ have two real roots in $(0, \pi)$, then $a \in$

(A) (1, 2) (B) (3, 4) (C) $(3, 4) \cup \{5\}$ (D) (3, 5)

TE0042

5. If $f(x) \geq 0 \forall x \in \mathbb{R}$ then range of 'a' is

(A) $[2, \infty)$ (B) $[4, \infty)$ (C) $(4, \infty)$ (D) none of these

TE0042

Paragraph for Questions 6 to 8

On the basis of the given polynomial equation

$$x^4 - (\sin\theta + \cos\theta)x^3 + (\sin\theta \cos\theta - 1)x^2 + (\sin\theta + \cos\theta)x - \sin\theta \cos\theta = 0.$$

Answer the following :

6. Sum of any 2 roots of the equation can never exceed -

(A) -1 (B) 0 (C) 1 (D) 2

TE0106

7. Product of the roots of the equation will always be more than -

(A) -1 (B) 0 (C) 1 (D) 2

TE0107

8. Number of real roots of the equation is/are -

(A) 1 (B) 2 (C) 3 (D) 4

TE0108

Matrix Match Type

9.	List-I	List-II	
(I)	Number of common solutions of the equations $2 \cos^2 x - 3 \cos x + 1 = 0$ and $\tan\left(\frac{3x}{4}\right) + 1 = 0$, where $-\pi < x \leq 3\pi$, is equal to	(P) 2	TE0043
(II)	Number of solutions of the equation $\sin^2 x + 2 \sin x - 4 \cos x - \sin 2x = 0$ in $(0, 2\pi)$ is equal to	(Q) 12	TE0044
(III)	If the sum of all possible values of $x \in (0, 2\pi)$ satisfying the equation $2 \cos x \operatorname{cosec} x - 4 \cos x - \operatorname{cosec} x = -2$ is equal to $\frac{k\pi}{4}$ ($k \in \mathbb{N}$), then the value of k is	(R) 4 (S) 3	TE0045
(IV)	Let $x - y = \frac{1}{4}$ and $\cos(\pi x) \cdot \cos(\pi y) = \frac{1}{2\sqrt{2}}$ where $x, y \in (0, 2)$ then the number of ordered pair(s) (x, y) is equal to	(T) 1	TE0046
(A) $I \rightarrow P; II \rightarrow P; III \rightarrow Q; IV \rightarrow R$ (B) $I \rightarrow T; II \rightarrow P; III \rightarrow S; IV \rightarrow R$ (C) $I \rightarrow R; II \rightarrow P; III \rightarrow S; IV \rightarrow T$ (D) $I \rightarrow Q; II \rightarrow P; III \rightarrow S; IV \rightarrow T$			
10.	Column I	Column II	
(A)	The number of real roots of the equation $\cos^7 x + \sin^4 x = 1$ in $(-\pi, \pi)$ is	(P) 1	TE0047
(B)	The number of solutions of the equation $ \cot x = \cot x + \frac{1}{\sin x} \quad (0 < x < \pi)$ is	(Q) 2	TE0048
(C)	The number of solution(s) of equation in $(0, 3\pi)$ $\sin^2 x + 3 \sec^2 x = -(\tan^2 x + \cot^2 x)$ is	(R) 0 (S) 3	TE0050
(D)	The number of values of $x \in [-2\pi, 2\pi]$, which satisfy $\operatorname{cosec} x = 1 + \cot x$	(T) 5	

EXERCISE (O-4)**Numerical Grid Type**

1. If $\sin \pi x + \cos \pi x = 0$, then the sum of all solutions in $[0, 100]$ is
TE0055
2. If α, β ($\alpha - \beta \neq 2n\pi, n \in \mathbb{I}$) are different values of θ satisfying the equation $5 \cos \theta - 12 \sin \theta = 11$. If the value of $\sin (\alpha + \beta) = -\frac{5k}{169}$, then the value of k is
TE0109
3. If $x \in \left(\pi, \frac{3\pi}{2}\right)$ then $4 \cos^2\left(\frac{\pi}{4} - \frac{x}{2}\right) + \sqrt{4 \sin^4 x + \sin^2 2x}$ is always equal to
TE0110
4. The number of integral values of λ , for which equation $4 \cos x + 3 \sin x = 2\lambda + 1$ has a solution is
TE0111
5. If $2 \cos x + \sin x = 1$, then find the sum of all possible values of $7 \cos x + 6 \sin x$.
TE0112
6. If $2 \tan^2 x - 5 \sec x - 1 = 0$ has 7 different roots in $\left[0, \frac{n\pi}{2}\right]$, $n \in \mathbb{N}$, then find the greatest value of n .
TE0113
7. If the arithmetic mean of the roots of the equation $4 \cos^3 x - 4 \cos^2 x - \cos(\pi + x) - 1 = 0$ in the interval $[0, 315]$ is equal to $k\pi$, then find the value of k
TE0114
8. In $(0, 6\pi)$, find the number of solutions of the equation $\tan \theta + \tan 2\theta + \tan 3\theta = \tan \theta \cdot \tan 2\theta \cdot \tan 3\theta$
TE0115
9. Find the number of solutions of the equation $\cos 6x + \tan^2 x + \cos 6x \cdot \tan^2 x = 1$ in the interval $[0, 2\pi]$.
TE0116
10. Find the number of solutions of $\sin \theta + 2 \sin 2\theta + 3 \sin 3\theta + 4 \sin 4\theta = 10$ in $(0, \pi)$.
TE0117

EXERCISE (JM)

1. Let A and B denote the statements

$$\mathbf{A} : \cos \alpha + \cos \beta + \cos \gamma = 0$$

$$\mathbf{B} : \sin \alpha + \sin \beta + \sin \gamma = 0$$

If $\cos(\beta - \gamma) + \cos(\gamma - \alpha) + \cos(\alpha - \beta) = -\frac{3}{2}$, then -

[AIEEE 2009]

- (1) Both **A** and **B** are true
(2) Both **A** and **B** are false
(3) **A** is true and **B** is false
(4) **A** is false and **B** is true

TE0071

2. The possible values of $\theta \in (0, \pi)$ such that $\sin(\theta) + \sin(4\theta) + \sin(7\theta) = 0$ are: [AIEEE 2011]

(1) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{4\pi}{9}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(2) $\frac{\pi}{4}, \frac{5\pi}{12}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

(3) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{35\pi}{36}$

(4) $\frac{2\pi}{9}, \frac{\pi}{4}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{8\pi}{9}$

TE0072

3. In a ΔPQR , if $3 \sin P + 4 \cos Q = 6$ and $4 \sin Q + 3 \cos P = 1$, then the angle R is equal to :

[AIEEE-2012]

(1) $\frac{3\pi}{4}$

(2) $\frac{5\pi}{6}$

(3) $\frac{\pi}{6}$

(4) $\frac{\pi}{4}$

TE0073

4. If $0 \leq x < 2\pi$, then the number of real values of x, which satisfy the equation

$$\cos x + \cos 2x + \cos 3x + \cos 4x = 0, \text{ is :-}$$

[JEE(Main) 2016]

(1) 9

(2) 3

(3) 5

(4) 7

TE0074

5. If sum of all the solutions of the equation $8 \cos x \cdot \left(\cos\left(\frac{\pi}{6} + x\right) \cdot \cos\left(\frac{\pi}{6} - x\right) - \frac{1}{2} \right) = 1$ in $[0, \pi]$ is $k\pi$, then k is equal to :

[JEE(Main) 2018]

(1) $\frac{13}{9}$

(2) $\frac{8}{9}$

(3) $\frac{20}{9}$

(4) $\frac{2}{3}$

TE0075

6. If $0 \leq x < \frac{\pi}{2}$, then the number of values of x for which $\sin x - \sin 2x + \sin 3x = 0$, is

[JEE(Main) 19]

(1) 2

(2) 1

(3) 3

(4) 4

TE0076

7. The sum of all values of $\theta \in \left(0, \frac{\pi}{2}\right)$ satisfying $\sin^2 2\theta + \cos^4 2\theta = \frac{3}{4}$ is : [JEE(Main) 19]

(1) $\frac{\pi}{2}$ (2) π (3) $\frac{3\pi}{8}$ (4) $\frac{5\pi}{4}$

TE0077

8. Let $S = \{\theta \in [-2\pi, 2\pi] : 2\cos^2 \theta + 3\sin \theta = 0\}$. Then the sum of the elements of S is [JEE(Main) 19]

(1) $\frac{13\pi}{6}$ (2) π (3) 2π (4) $\frac{5\pi}{3}$

TE0078

9. All the pairs (x, y) that satisfy the inequality $2\sqrt{\sin^2 x - 2\sin x + 5} \cdot \frac{1}{4^{\sin^2 y}} \leq 1$ also satisfy the equation.

[JEE(Main) 19]

(1) $\sin x = |\sin y|$ (2) $\sin x = 2 \sin y$ (3) $2|\sin x| = 3\sin y$ (4) $2\sin x = \sin y$

TE0079

10. The number of solutions of the equation $1 + \sin^4 x = \cos^2 3x$, $x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right]$ is : [JEE(Main) 19]

(1) 5 (2) 4 (3) 7 (4) 3

TE0080

11. Let S be the set of all $\alpha \in \mathbb{R}$ such that the equation, $\cos 2x + \alpha \sin x = 2\alpha - 7$ has a solution. Then S is equal to : [JEE(Main) 19]

(1) $[2, 6]$ (2) $[3, 7]$ (3) $\mathbb{R}$ (4) $[1, 4]$

TE0081

12. If $\sin^4 \alpha + 4\cos^4 \beta + 2 = 4\sqrt{2} \sin \alpha \cos \beta$; $\alpha, \beta \in [0, \pi]$, then $\cos(\alpha + \beta) - \cos(\alpha - \beta)$ is equal to : [JEE(Main) 2019]

(1) 0 (2) $-\sqrt{2}$ (3) -1 (4) $\sqrt{2}$

TE0082

13. The number of distinct solutions of the equation $\log_{\frac{1}{2}} |\sin x| = 2 - \log_{\frac{1}{2}} |\cos x|$ in the interval $[0, 2\pi]$, is ———. [JEE(Main) 20]

TE0083

14. All possible values of $\theta \in [0, 2\pi]$ for which $\sin 2\theta + \tan 2\theta > 0$ lie in : [JEE(Main) 2021]

(1) $\left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$

(2) $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

(3) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right)$

(4) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$

TE0118

15. The number of solutions of the equation $\cos\left(x + \frac{\pi}{3}\right)\cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4}\cos^2 2x$, $x \in [-3\pi, 3\pi]$ is :

(A) 8

(B) 5

(C) 6

(D) 7

[JEE(Main) 2022]

TE0119

EXERCISE (JA)

1. For $x \in (0, \pi)$, the equation $\sin x + 2\sin 2x - \sin 3x = 3$ has [JEE(Advanced)-2014, 3(-1)]
 (A) infinitely many solutions (B) three solutions
 (C) one solution (D) no solution

TE0087

2. The number of distinct solutions of equation $\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is [JEE 2015, 4M, -0M]
 TE0088

3. Let $S = \left\{ x \in (-\pi, \pi) : x \neq 0, \pm \frac{\pi}{2} \right\}$. The sum of all distinct solution of the equation $\sqrt{3} \sec x + \operatorname{cosec} x + 2(\tan x - \cot x) = 0$ in the set S is equal to -

[JEE(Advanced)-2016, 3(-1)]

- (A) $-\frac{7\pi}{9}$ (B) $-\frac{2\pi}{9}$ (C) 0 (D) $\frac{5\pi}{9}$

TE0089

4. Let a, b, c be three non-zero real numbers such that the equation

$$\sqrt{3}a \cos x + 2b \sin x = c, \quad x \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

has two distinct real roots α and β with $\alpha + \beta = \frac{\pi}{3}$. Then the value of $\frac{b}{a}$ is _____

[JEE(Advanced)-2018, 3(0)]

TE0090

Answer the following by appropriately matching the lists based on the information given in the paragraph

Let $f(x) = \sin(\pi \cos x)$ and $g(x) = \cos(2\pi \sin x)$ be two functions defined for $x > 0$. Define the following sets whose elements are written in the increasing order :

$$X = \{x : f(x) = 0\}, \quad Y = \{x : f'(x) = 0\}$$

$$Z = \{x : g(x) = 0\}, \quad W = \{x : g'(x) = 0\}.$$

List-I contains the sets X, Y, Z and W. List -II contains some information regarding these sets.

List-I

- (I) X
(II) Y
(III) Z
(IV) W

List-II

- (P) $\supseteq \left\{ \frac{\pi}{2}, \frac{3\pi}{2}, 4\pi, 7\pi \right\}$
(Q) an arithmetic progression
(R) NOT an arithmetic progression
(S) $\supseteq \left\{ \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6} \right\}$
(T) $\supseteq \left\{ \frac{\pi}{3}, \frac{2\pi}{3}, \pi \right\}$
(U) $\supseteq \left\{ \frac{\pi}{6}, \frac{3\pi}{4} \right\}$

5. Which of the following is the only CORRECT combination ? [JEE(Advanced)-2019, 3(-1)]
(1) (II), (R), (S) (2) (I), (P), (R) (3) (II), (Q), (T) (4) (I), (Q), (U)

TE0091

6. Which of the following is the only CORRECT combination ? [JEE(Advanced)-2019, 3(-1)]
(1) (IV), (Q), (T) (2) (IV), (P), (R), (S) (3) (III), (R), (U) (4) (III), (P), (Q), (U)

TE0091

7. Let $f : [0, 2] \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = (3 - \sin(2\pi x)) \sin \left(\pi x - \frac{\pi}{4} \right) - \sin \left(3\pi x + \frac{\pi}{4} \right)$$

[JEE(Advanced)-2020]

If $\alpha, \beta \in [0, 2]$ are such that $\{x \in [0, 2] : f(x) \geq 0\} = [\alpha, \beta]$, then the value of $\beta - \alpha$ is _____

TE0100

8. Consider the following lists :

List-I

- (I) $\left\{ x \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3} \right] : \cos x + \sin x = 1 \right\}$
(II) $\left\{ x \in \left[-\frac{5\pi}{18}, \frac{5\pi}{18} \right] : \sqrt{3} \tan 3x = 1 \right\}$
(III) $\left\{ x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5} \right] : 2 \cos(2x) = \sqrt{3} \right\}$
(IV) $\left\{ x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4} \right] : \sin x - \cos x = 1 \right\}$

List-II

- (P) has two elements
(Q) has three elements
(R) has four elements
(S) has five elements
(T) has six elements

The correct option is :

[JEE(Advanced)-2022]

- (A) (I) $\rightarrow$ (P); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (S)
(B) (I) $\rightarrow$ (P); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (R)
(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (P); (III) $\rightarrow$ (T); (IV) $\rightarrow$ (S)
(D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

TE0120

ANSWER KEY

Do yourself-1

1. (a) $\theta = n\pi + (-1)^n \frac{\pi}{6}, n \in I$ (b) $\theta = (2n+1)\frac{\pi}{3}, n \in I$ (c) $\theta = \frac{4n\pi}{3}, n \in I$
 (d) $\theta = \frac{n\pi}{2}, n \in I$ (e) $\theta = n\pi \pm \frac{\pi}{12}, n \in I$ (f) $\theta = 2n\pi + (-1)^{n+1}\pi, n \in I$
2. A, B, C, D
3. D
4. $\frac{n\pi}{4} \pm \sqrt{1 + \frac{n^2\pi^2}{16}}, n \in I$
5. (i) $n\pi + (-1)^n \frac{\pi}{4}, n \in I$ (ii) $n\pi + \frac{\pi}{3} + 1, n \in I$ (iii) $n\pi - \frac{\pi}{4}, n \in I$
 (iv) $n\pi + (-1)^n \frac{\pi}{3}, n \in I$ (v) $n\pi \pm \frac{\pi}{4}, n \in I$
6. (i) $\frac{m\pi}{4}, m \in I$ or $\frac{(2m+1)\pi}{10}, m \in I$ (ii) $2n\pi \pm \frac{\pi}{3}, n \in I$
 (iii) $\left(2n + \frac{1}{2}\right)\frac{\pi}{5}, n \in I$ or $2n\pi - \frac{\pi}{2}, n \in I$ (iv) $\left(n + \frac{1}{2}\right)\frac{\pi}{9}, n \in I$
 (v) $\left(n + \frac{1}{4}\right)\frac{\pi}{2}, n \in I$ (vi) $2n\pi + \frac{2\pi}{3}, n \in I$
 (vii) $n\pi \pm \frac{\pi}{6}$ (viii) $\left(n + \frac{1}{3}\right)\frac{\pi}{3}, n \in I$

Do yourself-2

1. (a) $x = n\pi + (-1)^{n+1} \frac{\pi}{6}, n \in I$ (b) $\alpha = \frac{n\pi}{2}$ or $\alpha = \frac{k\pi}{2} + \frac{3\pi}{8}, n, k \in I$
 (c) $\theta = n\pi \pm \frac{\pi}{3}, n \in I$
 (d) $\theta = n\pi + (-1)^n \alpha$, where $\alpha = \sin^{-1}\left(\frac{\sqrt{17}-1}{8}\right)$ or $\sin^{-1}\left(\frac{-1-\sqrt{17}}{8}\right), n \in I$
2. $\theta = \left\{-\frac{\pi}{4}, -\frac{3\pi}{4}, -\frac{\pi}{2}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{2}\right\}$
3. $(2n+1)\pi, n \in I$ 4. $2n\pi - \frac{\pi}{2}, n \in I$ or $n\pi + (-1)^{n+1} \frac{\pi}{6}, n \in I$
5. B 6. 3 7. B
8. (i) $n\pi + \frac{\pi}{3}, n \in I$ or $n\pi + \frac{\pi}{4}, n \in I$
 (ii) $n\pi + (-1)^n \frac{\pi}{10}, n \in I$ or $n\pi - (-1)^n \frac{3\pi}{10}, n \in I$
 (iii) $x = \frac{n\pi}{3} - \frac{\pi}{9}, n \in I$

Do yourself-3

1. (a) $x = 2n\pi - \frac{\pi}{4}, n \in I$ (b) $2m\pi + \frac{\pi}{2}, m \in I$
2. $2n\pi, n \in I$ or $\frac{2n\pi}{3} + \frac{\pi}{6}, n \in I$ 3. $x = n\pi + \frac{\pi}{6}, n \in I, x = \frac{n\pi}{2} - \frac{\pi}{12}, n \in I$
4. (i) $n\pi + \frac{\pi}{6} + (-1)^n \frac{\pi}{4}, n \in I$ (ii) $2n\pi + \frac{\pi}{2}, n \in I$ or $2n\pi + 2\alpha$ where $\alpha = \tan^{-1} \frac{3}{7}, n \in I$

Do yourself-4

1. $\theta = n\pi$ or $\theta = \frac{m\pi}{3} \pm \frac{\pi}{9}; n, m \in I$ 2. $x = \frac{n\pi}{3}, n \in I$ and $k\pi \pm \frac{\pi}{3}, k \in I$
3. $n\pi - \frac{\pi}{4}, n \in I$ 4. $n\pi - \frac{\pi}{4}, n \in I$
5. $2n\pi + \frac{\pi}{2}, n \in I$ or $2n\pi, n \in I$ or $n\pi + \frac{\pi}{4}, n \in I$
6. D
7. (i) $2n\pi, n \in I$ or $\frac{2n\pi}{3} + \frac{\pi}{6}, n \in I$
- (ii) $x = (2n+1)\frac{\pi}{4}, n \in I$ or $x = (2n+1)\frac{\pi}{2}, n \in I$ or $x = n\pi \pm \frac{\pi}{6}, n \in I$
- (iii) $m\pi, m \in I$ or $\frac{m\pi}{n-1}, m \in I$ or $\left(m + \frac{1}{2}\right)\frac{\pi}{n}, m \in I$

Do yourself-5

1. D 2. $x = \frac{\pi}{4}$ 3. $(4p-3)\frac{\pi}{2}, p \in I$ 4. $2m\pi + \frac{\pi}{2}, m \in I$
5. A 6. C

Do yourself - 6

1. $\bigcup_{n \in I} \left[2n\pi - \frac{2\pi}{3}, 2n\pi + \frac{2\pi}{3} \right]$ 2. $\left[\frac{\pi}{6}, \frac{5\pi}{6} \right]$ 3. $x \in \left[n\pi - \frac{\pi}{4}, n\pi + \frac{\pi}{4} \right] : n \in I$
4. $\left(2n\pi + \frac{7\pi}{6}, 2n\pi + \frac{11\pi}{6} \right)$ 5. $\theta \in (n\pi + \pi/3, n\pi + \pi/2]$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	D	C	C	C	B	A	C	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	C	D	C	C	D	B	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	D	A	C	A	B	D	A	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	B,C	A,B	B,C	A,C	A,B,C,D	B,C	A,C,D	A,C	B,C
Que.	11	12	13	14	15					
Ans.	B,D	B,D	A,B,C,D	A,B,C	A,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	C	C	A	B	B	D	A	D	A	
Q.10	A	B	C	D						
	S	P	R	Q						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	5025	24	2	6	8	15.00	50.00	17.00	7.00	0.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	3	4	1	1	1	3	1	1
Que.	11	12	13	14	15					
Ans.	1	2	8.00	4	D					

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8		
Ans.	D	8	C	0.50	3	2	1.00	B		